

AI BASED DOS DETECTION FOR SOC

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THE PROBLEM

1. DoS attacks can disrupt the availability of network services.
2. SOC analysts need to detect attacks quickly while minimizing false alarms.
3. A simple binary prediction is not enough: analysts also need context, confidence, and supporting evidence.

Goal: Support faster and more informed SOC decisions.

PROJECT GOAL

Build an AI assisted SOC pipeline that can:

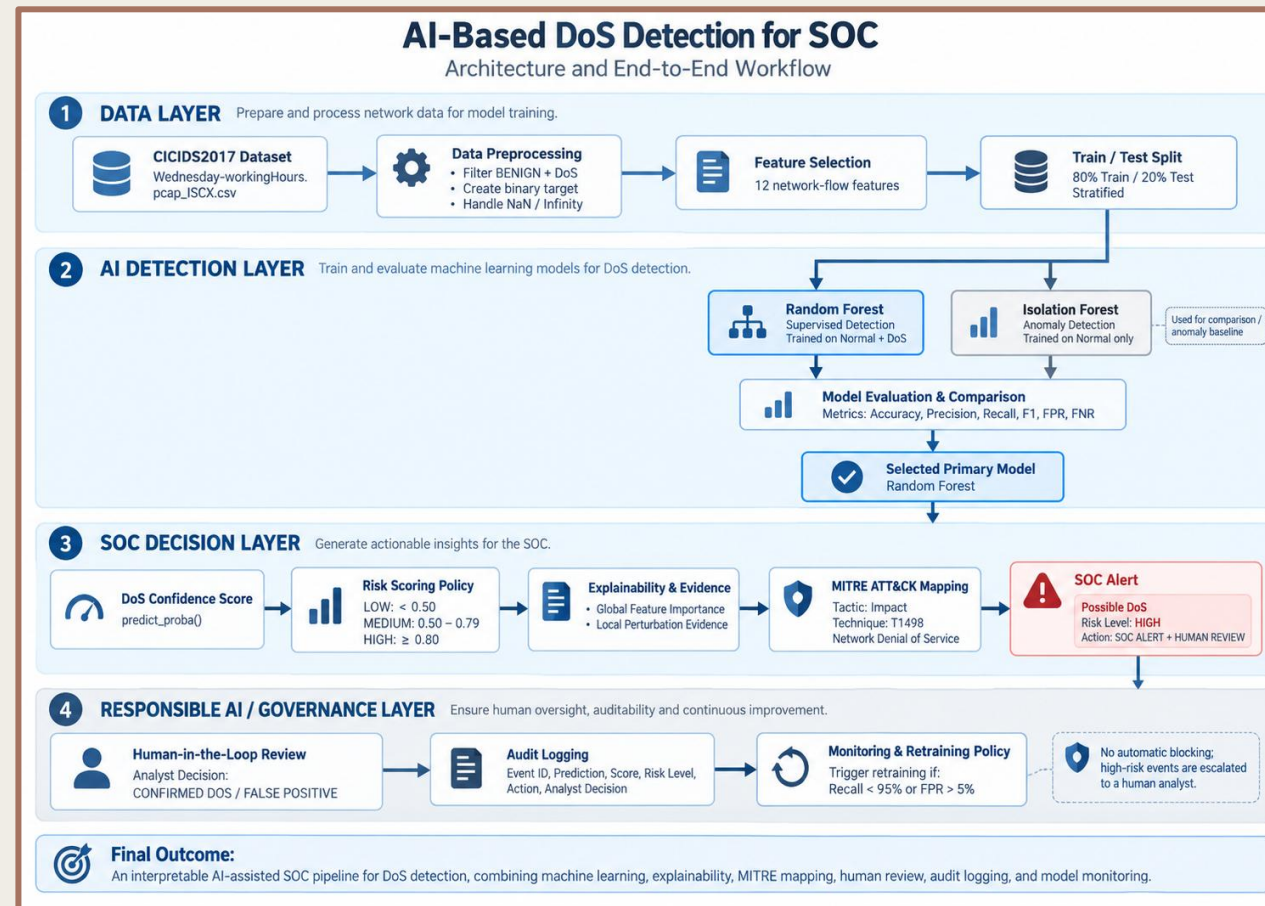
- Detect Normal vs. DoS network traffic
- Compare supervised and anomaly detection approaches
- Generate a DoS confidence score and risk level
- Provide explainable evidence
- Support human review and audit logging

The system does not automatically block traffic.

DATASET – CICIDS2017

- Wednesday Working Hours dataset
- Contains benign traffic and several DoS attack types
- Original labels include:
BENIGN, DoS Hulk, DoS GoldenEye, DoS slowloris, DoS Slowhttptest
- Final task: **Normal vs. DoS**

SYSTEM ARCHITECTURE



End-to-end pipeline from raw network traffic to SOC alert, human review, logging, and monitoring.

DATA PREPARATION & FEATURE SELECTION

Preprocessing

- Keep BENIGN and DoS traffic
- Binary target: 0 = Normal, 1 = DoS
- Handle missing and infinite values
- 80/20 stratified train-test split

12 Selected Network-Flow Features

- Flow Duration
- Packet counts and lengths
- Flow Bytes/s
- Flow Packets/s
- SYN / ACK flags

MACHINE LEARNING MODELS

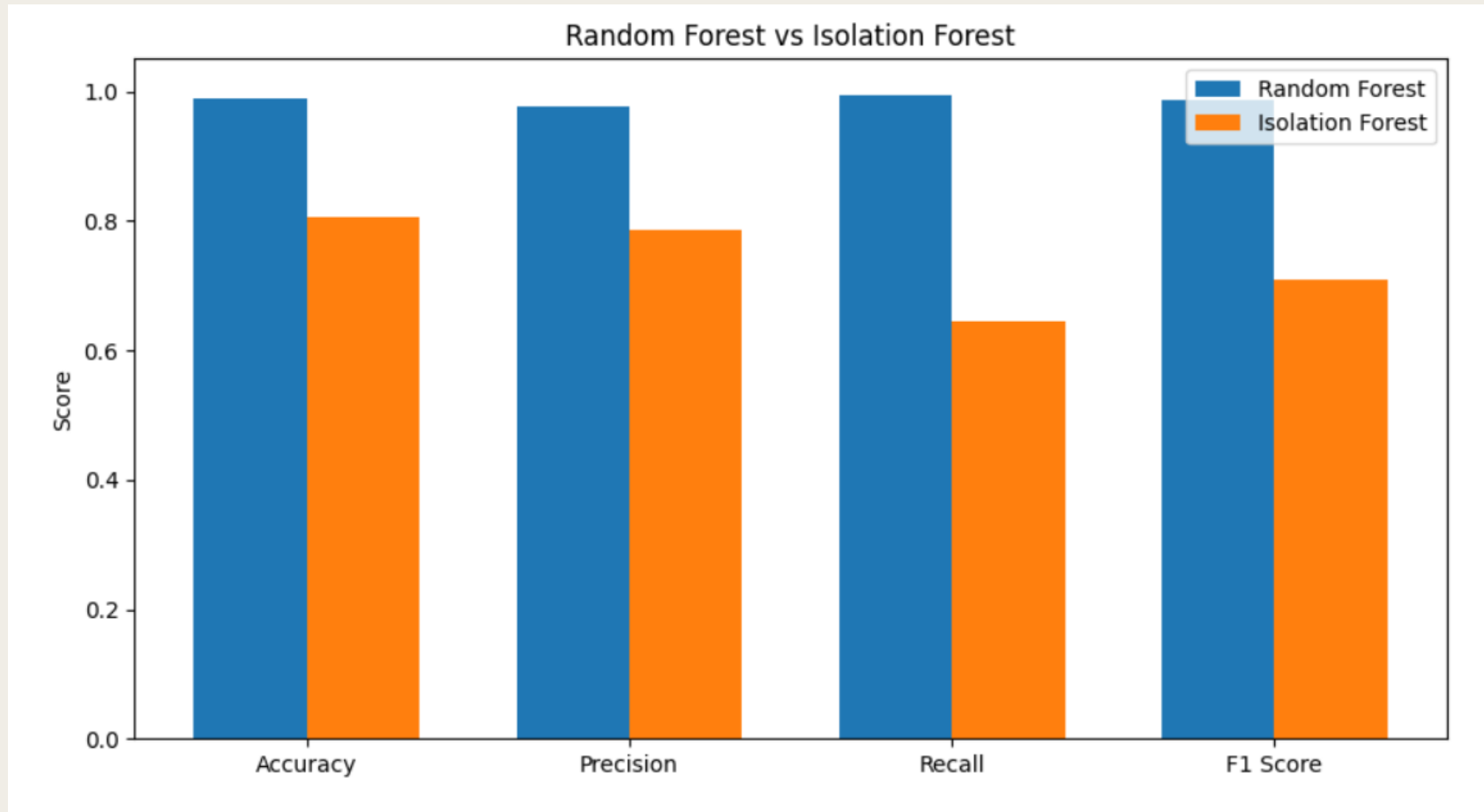
Random Forest

- Supervised learning
- Trained on Normal + DoS traffic
- Learns known DoS patterns

Isolation Forest

- Anomaly detection
- Trained only on Normal traffic
- Detects unusual network behaviour

MODEL PERFORMANCE



Random Forest

- Accuracy: **98.96%**
- Precision: **97.73%**
- Recall: **99.45%**
- F1 Score: **98.58%**

Random Forest achieved consistently stronger performance.

SECURITY RELEVANT ERROR RATES

Random Forest

False Positive Rate: **1.32%**

False Negative Rate: **0.55%**

Isolation Forest

False Positive Rate: **10.08%**

False Negative Rate: **35.50%**

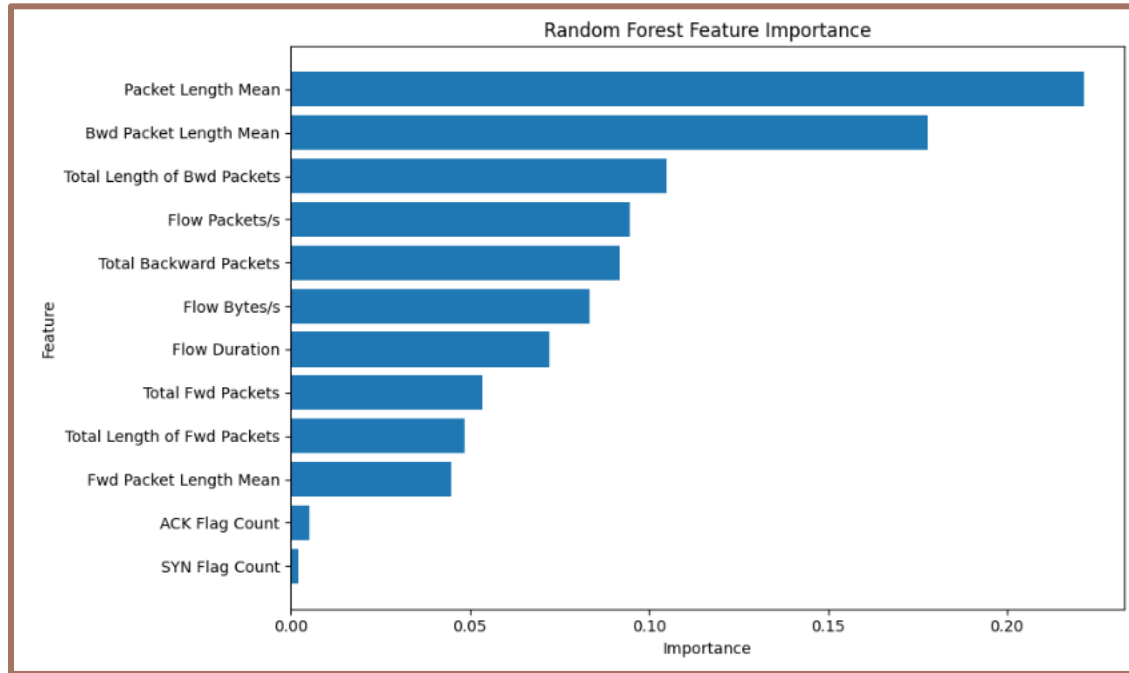
Random Forest missed far fewer DoS attacks and produced fewer false alarms.

SOC RISK SCORING & POLICY GATE



No automatic blocking is performed.

EXPLAINABILITY & HUMAN IN THE LOOP



```
analyst_review(6, "CONFIRMED DOS")
```

Analyst decision saved.

Event ID	Prediction	DoS Score	Risk Level	SOC Action
5	6	1	0.966002	HIGH

SOC ALERT + HUMAN REVIEW

Analyst Decision

5	CONFIRMED DOS
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Human Review

- High-risk alerts are escalated to an analyst
- Analyst can confirm DoS or mark a false positive

Global Explainability

- Random Forest feature importance
- Most influential features include Packet Length Mean and Bwd Packet Length Mean

**AI supports the analyst,
it does not replace the analyst.**

FINAL SOC ALERT

MITRE ATT&CK

1. Tactic: Impact
2. Technique: T1498
3. Network Denial of Service

The final alert combines detection, risk, evidence, MITRE mapping, and analyst decision in one output.

```
===== SOC ALERT =====  
Event ID: 6  
Prediction: Possible DoS  
DoS Score: 0.966  
Risk Level: HIGH  
  
MITRE ATT&CK:  
Tactic: Impact  
Technique ID: T1498  
Technique: Network Denial of Service  
  
SOC Action:  
SOC ALERT + HUMAN REVIEW  
  
Analyst Decision:  
CONFIRMED DOS  
  
Top Local Evidence:  


|   | Feature                | Impact   |
|---|------------------------|----------|
| 9 | ACK Flag Count         | 0.966002 |
| 6 | Fwd Packet Length Mean | 0.379658 |
| 8 | Packet Length Mean     | 0.371497 |

  
=====
```

MONITORING & RETRAINING POLICY

Retraining is recommended if:

Recall < 95%

False Positive Rate > 5%

Current Random Forest Performance

Recall: 99.45%

FPR: 1.32%

MODEL PERFORMANCE STABLE

In this prototype, the policy is demonstrated using test-set performance.

CONCLUSION

- Random Forest achieved the best results for the selected DoS detection task
- The final pipeline goes beyond classification by adding risk scoring and explainability
- MITRE ATT&CK provides standardized security context
- Human-in-the-Loop prevents fully automated security decisions
- Audit logging and monitoring improve traceability and future maintainability

AI assisted detection can support faster, more explainable, and more responsible SOC decision making.